

BACKLOG BUDDY ? STUDY MATERIAL

Engineering Mathematics-I ? Syllabus & Reference Books

Subject Code: MA101 Semester: 1

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Unit-wise syllabus, recommended reference books and question paper pattern.

Unit 1: Matrices and Linear Algebra

- Definition, key concepts and significance of Matrices and Linear Algebra
- Important diagrams / derivations / algorithms for Matrices and Linear Algebra
- Solved examples and previous-year questions on Matrices and Linear Algebra

Unit 2: Differential Calculus

- Definition, key concepts and significance of Differential Calculus
- Important diagrams / derivations / algorithms for Differential Calculus
- Solved examples and previous-year questions on Differential Calculus

Unit 3: Integral Calculus

- Definition, key concepts and significance of Integral Calculus
- Important diagrams / derivations / algorithms for Integral Calculus
- Solved examples and previous-year questions on Integral Calculus

Unit 4: Vector Calculus

- Definition, key concepts and significance of Vector Calculus
- Important diagrams / derivations / algorithms for Vector Calculus
- Solved examples and previous-year questions on Vector Calculus

Unit 5: Ordinary Differential Equations

- Definition, key concepts and significance of Ordinary Differential Equations
- Important diagrams / derivations / algorithms for Ordinary Differential Equations
- Solved examples and previous-year questions on Ordinary Differential Equations

Exam Tips:

- * Start with the latest syllabus and exam pattern ? focus on units with the highest weightage first.
- * Solve at least 3 previous year papers under timed conditions before the exam.
- * Focus on frequently repeated topics ? they are strong preparation priorities based on available papers (not guarantees).
- * Revise definitions, diagrams and standard derivations ? they earn easy marks.
- * Practise numerical problems from every unit; papers usually include 2-3 compulsory problems.

Disclaimer: Sample study material for preparation support only.

Appearance counts are based on available papers and are not exam predictions.